



Figure 2: Unsupervised learning using clustering

As per the graph given in Figure 2, let's consider a company that has only the shoulder and waist measurements as input parameters of the data set. It will now have to categorise this data set into three groups, which can help the company predict the shirt size for every customer. This technique is known as clustering, in which the data set is clustered into the desired number of clusters. Most of the time, the data set is not like the one shown in this example. Data points that are very close to each other make it difficult to implement clustering. Also, clustering is just one of the many techniques used in unsupervised learning to predict the outcome.

3. **Reinforcement learning:** In reinforcement learning, a machine or an agent trains itself when exposed to a particular environment, by a process of trial and error. Let's consider a kid who wants to learn to ride a bicycle. First, she will try to learn from someone who already knows how to ride a bicycle. Then, she will try riding on her own and might fall down a number of times. Learning from her previous mistakes, she will try to ride without falling. And when she finally rides the bike without falling, it can be considered as a reward for her efforts. Now let's consider this kid as a machine or an agent who is getting punished (falling) for committing a mistake and earning a reward (not falling) for not committing any mistake. A chess-playing program can be a good example of this, whereby one wrong move will penalise the agent and it may lose a game, while a combination of one or more right moves will earn it a reward by making it win.

These are the three basic learning models of machine learning. As per the need, these models can be used in combination to generate a new model. For instance, supervised learning can sometimes be used along with unsupervised learning, depending upon the data set as well as the expected outcome.

People often feel that machine learning is only for someone who is good with mathematics or statistics, and will be impossible to learn for anyone else. This is a fallacy. Machine learning is not rocket science after all. The only thing that is required to learn it is eagerness and curiosity. The number of tools and libraries available has made it even easier to learn it. Google's TensorFlow library, which is now open source, or the numerous Python libraries like

NumPy and scikit-learn, are just a few of these. Anyone can use these libraries and even contribute to them to solve problems, as they are open source. You don't need to worry about the complexities involved in your algorithm, like complex mathematical computations (such as gradient, matrix multiplication, etc) as this task can be left for these libraries to implement. Libraries make it easier for everyone so that instead of getting involved in implementing complex computations, the user can now focus on the application of the algorithm.

There are also many APIs available that can be used to implement an AI based system. One such API is IBM's Watson – a cognitive, computing based AI system. Cognitive computing is a mixture of different techniques such as machine learning, natural language processing, AI, etc. Watson is capable of doing many tasks like answering a user's queries, helping doctors to spot diseases, and a lot more.

If you are excited by the prospects that machine learning offers, our digital education era has made things easier for you. There are many massive open online courses (MOOC) offered by many companies. One such course is provided by Coursera-Machine Learning. This is taught by Andrew Ng, one of the co-founders of Coursera. This course will give you a basic understanding of the algorithms that are implemented in machine learning, and it includes both supervised learning and unsupervised learning. It's a self-paced course but designed to be completed in 12 weeks. If you want to dig deeper and study deep learning, which is a subset of machine learning, you can learn it through another course provided by *fast.ai*. This course is split into two parts: *Practical deep learning for coders (Part 1)* and *Cutting edge deep learning for coders (Part 2)*. Both have been designed for seven weeks each and provide you a great insight into deep learning. If you further want to specialise in deep learning, you can opt for a deep learning specialisation course by Coursera and *deeplearning.ai*.

We all know that theory without practice is like a body without life. So, for you to practice, there are many sources that can provide you a huge data set to test your knowledge and implement what you've learnt. One such website is *Kaggle*, which provides a diverse data set and can help you overcome your major hurdle, i.e., getting data to test your learning model.

If you sometimes feel lost in this journey of learning, when your algorithm does not work as expected or when you don't understand a complex equation, remember the famous dialogue in the movie, *The Pursuit of Happyness*: "Don't ever let someone tell you that you can't do something. Not even me. You got a dream; you gotta protect it. When people can't do something themselves, they're gonna tell you that you can't do it. You want something, go get it. Period." 

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